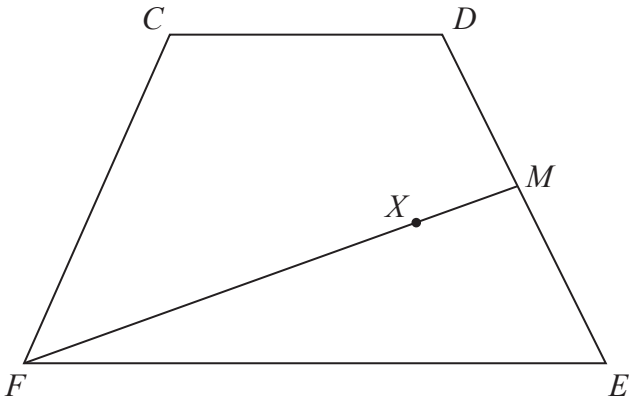
20 *CDEF* is a quadrilateral.



$\overrightarrow{CD} = \mathbf{a}$, $\overrightarrow{DE} = \mathbf{b}$ and $\overrightarrow{FC} = \mathbf{a} - \mathbf{b}$.

(a) Express $\overrightarrow{FE}$ in terms of **a** and/or **b**.
Give your answer in its simplest form.

F→E = F→C + C→D + D→E

= (a − b) + a + b = 2a

1 mark
Route F→C→D→E.

1 mark
2a.

2a

(2)

M is the midpoint of *DE*.
X is the point on *FM* such that $FX : XM = n : 1$
CXE is a straight line.

(b) Work out the value of *n*.

Take F as origin: C = a − b, E = 2a.

D = 2a − b, M = 2a − 1/2 b.

1 mark
Find M or a suitable route vector.

FX : XM = n : 1, so X = n/(n+1) M.

X = [2n/(n+1)]a − [n/2(n+1)]b.

1 mark
Express X using FM.

Since X lies on CE: X = (a − b) + t(a + b).

Equate coefficients: 2n/(n+1) = 1 + t

and −n/[2(n+1)] = −1 + t.

1 mark
Equate a, b coefficients.

Eliminate t: 5n = 4(n + 1).

n = 4

1 mark
n = 4.

$n =$ 4

(4)

(Total for Question 20 is 6 marks)